

Evidence-Based Assessment of the Unitree L1 Light Detection and Ranging Sensor for Indoor Mobile Robotics

Targeted literature review and secondary analysis of published performance data

Pascual Asio Serrano

Research internship - Institute of Systems and Robotics, University of Coimbra

Human-Centered Mobile Robotics Laboratory

6 August 2026

Abstract

This study asks under which operating and processing conditions the Unitree L1 light detection and ranging (LiDAR) sensor provides spatial information sufficiently stable for indoor robotic perception and mapping. A targeted review of manufacturer documentation, peer-reviewed field studies, a public dataset, and metrology guidance is combined with a secondary analysis of published odometry results. Unitree specifies a 360×90 degree field of view, 21,600 effective points per second, an 11 Hz horizontal scan rate, a 250 Hz inertial measurement unit (IMU) output, and ± 2 cm measurement accuracy. No independent controlled study was found that validates the ± 2 cm claim across distance, reflectivity, and incidence angle. In six environments at the European Organization for Nuclear Research (CERN), a method designed for non-repetitive scans achieved an unweighted mean trajectory root-mean-square error (RMSE) of 0.261 m relative to a Point-LIO LiDAR-inertial reference. The corresponding means were 3.532 m for Small generalized iterative closest point and 2.839 m for Keep It Small and Simple iterative closest point. Independent cave mapping produced coherent large-scale morphology, but insufficient point density for detailed floor reconstruction. An indoor navigation study reached 86.5% of planned waypoints over five runs, while cumulative map drift caused major failures in two runs. The evidence supports navigation-scale mapping at low or moderate motion when LiDAR-inertial fusion, scan accumulation, or registration adapted to non-repetitive sampling is used. It does not support treating the L1 as a calibrated centimetric survey instrument, relying on conventional scan-to-scan registration in low-overlap scenes, or assuming detailed reconstruction and long-duration Robot Operating System 2 stability without direct testing.

Keywords: Unitree L1; light detection and ranging; mobile robotics; non-repetitive scanning; odometry; mapping; point cloud.

1. Introduction

Low-cost three-dimensional LiDAR sensors broaden access to mobile-robot perception, but a manufacturer specification does not by itself establish task-level reliability. The Unitree L1 combines a hemispherical scan pattern with a low point rate relative to many multi-beam sensors. Its non-repetitive trajectory increases spatial coverage over time, but also reduces the correspondence between consecutive point clouds. This distinction is central: a sensor may measure individual ranges accurately while still producing unstable odometry when overlap, timing, or scene geometry is inadequate.

The research question is therefore: under which conditions does the Unitree L1 provide measurements and spatial coverage sufficiently stable for indoor perception and mapping on a mobile robot? The analysis separates four quantities that are often conflated: single-point range accuracy, spatial sampling density, trajectory accuracy, and geometric map detail. The aim is not to replace direct experiments, but to determine what is already supported by public evidence and what remains scientifically unverified.

2. Materials and methods

2.1 Evidence identification

A targeted evidence search was completed on 6 August 2026. Search terms combined "Unitree L1" or "Unitree 4D LiDAR" with accuracy, performance, mapping, odometry, point density, navigation, and dataset. Sources were retained when they identified the L1 explicitly and provided quantitative specifications, measured outcomes, raw data, or direct field observations relevant to the research question. The evidence base comprises the Unitree user manual [1], a six-environment CERN odometry study and its public dataset [2], [3], a lava-tube mapping study [4], an indoor quadruped navigation study [5], a review of terrestrial laser-scanner metrology [6], and the Point-LIO method used as the fused trajectory reference in the CERN study [7]. Studies that merely listed the sensor, lacked accessible performance results, or evaluated an unrelated algorithm without sensor-level interpretation were not used for quantitative conclusions.

This work is a targeted review and secondary analysis, not a systematic review or a new experiment. It reports no new measurements from the Coimbra sensor. Manufacturer statements, independent observations, derived calculations, and scientific inferences are identified separately throughout the text.

2.2 Data extraction and secondary calculations

Environment-level trajectory RMSE and processing times were transcribed from Table 3 of Habibiroudkenar et al. [2]. For each method, the arithmetic mean and median were calculated across the six environments. The relative reduction in mean error was calculated as $100 \times (1 - \text{mean RMSE of the evaluated method} / \text{mean RMSE of the baseline})$. Each environment was weighted equally because sequence durations and trajectory-sample counts were not reported. The resulting statistics are descriptive and do not constitute a formal meta-analysis.

The nominal duration of one 11 Hz horizontal cycle is 90.9 ms. This value is used only as an indicative timing reference, because the published work does not establish that one processed point cloud corresponds exactly to one complete horizontal cycle. Point counts per cycle were derived independently from the manual: 43,200 raw samples per second correspond to approximately 3,927 raw samples per cycle, whereas 21,600 effective points per second correspond to approximately 1,964 effective points per cycle.

2.3 Interpretation criteria

Evidence is interpreted at the level directly supported by each source. A manufacturer specification is treated as a declared operating characteristic. A field result is treated as evidence for the tested system and environment, not as an isolated sensor property. A trajectory error relative to Point-LIO is treated as agreement with a LiDAR-inertial reference, not as absolute metrological ground truth. A conclusion is classified as supported only when the evidence directly addresses the corresponding quantity.

3. Results

3.1 Declared sensor performance and metrological status

Table 1. Principal Unitree L1 specifications and their evidential status.

Quantity	Published value	Evidence-based interpretation
Maximum range	Precision Measurement (PM): 20 m at 90% reflectivity and 10 m at 10%; Remote Measurement (RM): 30 m at 90% and 15 m at 10% [1]	Usable range depends materially on target reflectivity.
Measurement accuracy	+/- 2.0 cm; stated test condition: 25 degrees C, 90% reflectivity, 15 m [1]	Manufacturer specification; no independent controlled uncertainty study was found.
Near blind zone	0.05 m [1]	The manual states that accuracy near this limit is not guaranteed.
Field of view	360 x 90 degrees [1]	Broad hemispherical coverage; density is explicitly non-uniform and higher near the central region.
Point rate	43,200 raw samples/s; 21,600 effective points/s [1]	Raw sampling and effective output must not be treated as interchangeable.
Scan frequencies	11 Hz horizontal; 180 Hz vertical [1]	The non-repetitive pattern develops coverage over time rather than producing identical frames.
Inertial output	1 kHz internal sampling; 250 Hz reporting [1]	Supports LiDAR-inertial odometry, but public timing and bias-stability data are absent.

The +/- 2 cm specification cannot be converted into a general uncertainty statement. The manual reports one test condition and warns that reflectivity, temperature, vibration, fog, and close-range operation can degrade performance [1]. Established scanner-evaluation practice instead varies distance, target reflectivity, incidence angle, and viewing direction against a traceable reference [6]. No public L1 study meeting those conditions was identified. The scientifically justified conclusion is therefore that centimetric range accuracy is declared but not independently validated across the operating envelope.

3.2 Non-repetitive sampling and spatial coverage

The manual states that point density varies across the field of view and is higher near the center [1]. Consecutive point clouds therefore contain irregular and only partly overlapping samples. This property is beneficial for progressive static coverage, but it weakens conventional frame-to-frame correspondence during motion.

A reporting ambiguity is visible across the literature. Hidding et al. describe approximately 3,927 points per measurement [4]. This number equals 43,200 raw samples per second divided by the 11 Hz horizontal rate. The manual, however, distinguishes 43,200 raw samples from 21,600 effective points per second, which yields approximately 1,964 effective points per horizontal cycle [1]. The term "measurement" in the field study therefore appears to refer to raw samples or a cycle-level packet, not necessarily to valid effective returns. Experimental reports should count the points actually delivered in each message and should report invalid or discarded returns rather than infer density from nominal frequencies.

The lava-tube mission used stationary acquisitions, robot rotations, and scanning positions usually separated by 1 to 3 m; battery constraints reduced the final density to approximately one acquisition every 2 to 4 m [4]. The complete reconstruction captured the large-scale cave morphology, but the 656,889-point floor subset did not resolve detailed floor conditions. The authors therefore used photogrammetry for terrain characterization. This result supports deliberate accumulation or pauses when local detail matters, while also showing that broad coverage does not imply fine geometric resolution.

3.3 Odometry in the CERN environments

The strongest L1-specific quantitative evidence is provided by six CERN sequences containing tunnels, repetitive structures, dynamic clutter, and a symmetric dome [2], [3]. The evaluated spatial-context method adapts generalized iterative closest point registration to non-repetitive scans. It is compared with Small generalized iterative closest point (Small GICP) and Keep It Small and Simple iterative closest point (KISS-ICP). The reference trajectory is generated by Point-LIO, a tightly coupled LiDAR-inertial odometry method using the same L1 point cloud and its IMU [2], [7]. The reference is operationally useful, but it is not independent motion capture or survey-grade ground truth.

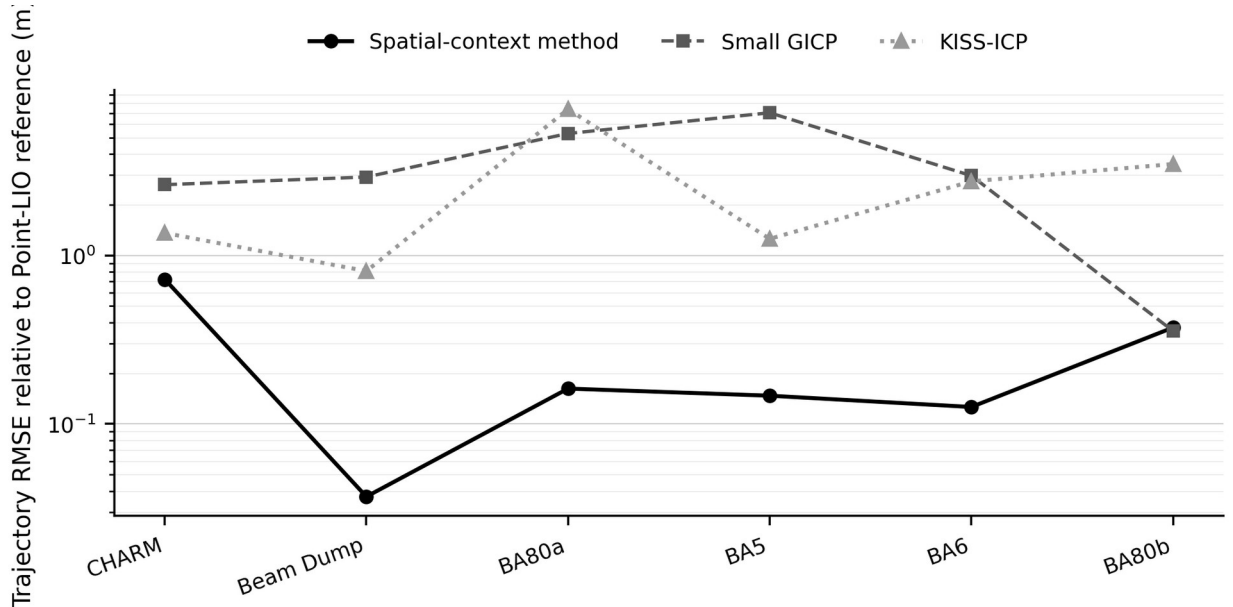


Figure 1. Published trajectory root-mean-square error for six Unitree L1 sequences [2]. The logarithmic vertical axis shows both successful and failure regimes. Errors are relative to a Point-LIO LiDAR-inertial reference, not to external metrology.

Table 2. Secondary summary of the CERN odometry results reported in reference [2].

Metric	Spatial-context method	Small GICP	KISS-ICP
Mean RMSE across six environments (m)	0.261	3.532	2.839
Median RMSE (m)	0.154	2.946	2.055
Environments with lower RMSE than spatial-context method	-	1 of 6	0 of 6
Mean reported processing time (s/scan)	0.131	0.094	0.00045
Evaluation mode	Offline	Offline comparison	Offline comparison

The spatial-context method achieved a mean RMSE of 0.261 m and a median of 0.154 m. The mean is 92.6% lower than Small GICP and 90.8% lower than KISS-ICP. It produced the lowest error in five of six comparisons against Small GICP and in all six comparisons against KISS-ICP. The exception is BA80b, where Small GICP reached 0.356 m and the spatial-context method 0.373 m. This exception prevents a claim of universal superiority and indicates that scene structure can occasionally favor a simpler registration method.

In the BA6 ablation, the spatial-context method reached 0.126 m RMSE, compared with 0.349 m for kinematic iterative closest point registration and 0.346 m for wheel odometry [2]. The reductions are 63.9% and 63.6%, respectively. These results show that scan-pattern-aware processing can recover useful odometry from the L1. They do not demonstrate centimetric absolute trajectory accuracy because the comparison remains tied to the Point-LIO reference.

3.4 Application-level mapping and navigation

The cave study illustrates the difference between local registration residual and global map agreement. Pairwise iterative closest point logs ranged from 0.025 to 0.091 m, while mean cloud-to-cloud distances ranged from 0.065 to 1.673 m [4]. A small optimizer residual can therefore coexist with a substantially misaligned reconstructed surface. The original online navigation map accumulated approximately 3.5 m of closure error when the robot returned to the cave entrance, after which scans were cleaned, manually aligned, and registered offline [4]. The result is evidence of a complete system limitation, not a direct measurement of L1 range error.

A separate indoor study used the L1 on a Unitree Go2 quadruped with Robot Operating System 2 (ROS 2), a two-dimensional simultaneous localization and mapping (SLAM) pipeline, and autonomous waypoint navigation [5]. Across five runs, the robot reached 86.5% of planned waypoints. Three runs reached every waypoint, whereas map drift made substantial portions of the route unreachable in two runs. This finding confirms that the L1 can support functional indoor navigation, but also that successful sensing does not prevent cumulative localization failure in a relative-sensor system.

3.5 Computational timing and stream stability

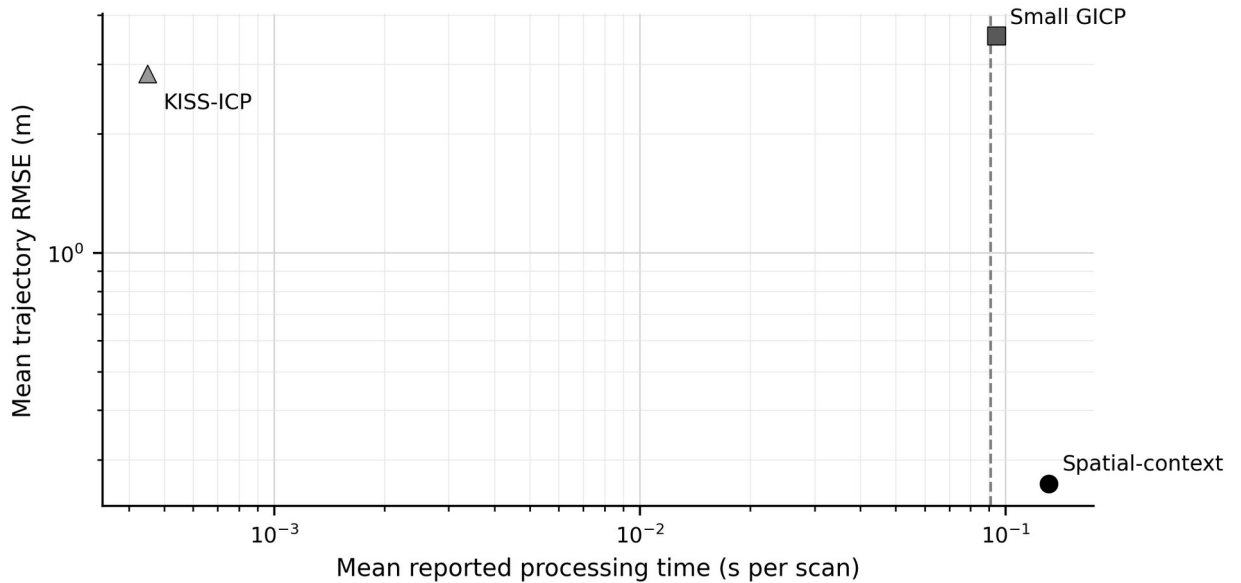


Figure 2. Accuracy-runtime trade-off derived from the six CERN environments [2]. Each point is an unweighted mean. The 90.9 ms line is the nominal period of the 11 Hz horizontal cycle; it is an indicative reference rather than a verified real-time deadline.

The spatial-context method required 0.131 s per scan on average, equivalent to approximately 7.6 processed scans per second. Five of six reported times exceeded 90.9 ms. Because the study explicitly used offline processing and did not define the exact relationship between a processed cloud and the 11 Hz mechanical cycle, the evidence does not establish end-to-end real-time operation. It only shows an implementation-level trade-off: the tailored method was much more accurate in these sequences, whereas KISS-ICP was far faster but often inaccurate.

No controlled long-duration L1 report was found for point-cloud and IMU frequency, timestamp jitter, packet loss, invalid returns, processor load, or thermal interruptions under ROS 2. Algorithm runtime and sensor-stream stability are different quantities. Public application studies demonstrate interoperability, but they do not replace a dedicated stability test.

4. Discussion

4.1 Direct answer to the research question

The published evidence supports a conditional answer. The Unitree L1 provides spatial information adequate for indoor navigation and coarse three-dimensional mapping when platform motion is low or moderate, the environment supplies repeated geometric constraints, and processing compensates for the non-repetitive scan through LiDAR-inertial fusion, temporal accumulation, or scan-context-aware registration. Stationary scans or deliberate pauses improve local

coverage when geometric detail is required. Operation should remain within the reflectivity-dependent range and thermal envelope declared by the manufacturer.

The evidence does not support four stronger uses. First, the L1 is not independently validated as a general ± 2 cm metrology instrument. Second, raw consecutive scans should not be assumed to work reliably with conventional iterative closest point registration in sparse, repetitive, symmetric, or low-overlap scenes. Third, the available density is not sufficient for detailed floor roughness or fine-surface reconstruction under the tested field acquisition. Fourth, guaranteed real-time mapping and long-duration ROS 2 stream stability remain unverified.

4.2 Why the published results are not contradictory

The literature appears inconsistent only when different performance quantities are treated as equivalent. The manufacturer accuracy specification concerns individual range measurements under stated conditions. Point rate and field of view concern how measurements are distributed in space and time. Trajectory RMSE concerns accumulated pose estimation relative to a reference trajectory. Map detail concerns whether the final sampling density resolves the geometry needed by the task. A sensor can satisfy one of these criteria and fail another. The L1 evidence is coherent when interpreted at the correct level: broad and useful navigation coverage is compatible with limited frame-to-frame overlap, cumulative drift, and insufficient fine-surface density.

4.3 Experiments that remain necessary

Three direct experiments remain scientifically non-redundant. A planar-target study should measure bias, repeatability, and point-to-plane dispersion against a calibrated reference while varying distance, reflectivity, and incidence angle. A static coverage study should quantify valid point count and occupied angular cells as a function of accumulation time. A ROS 2 stability test should record message rate, timestamp jitter, dropouts, invalid points, processor use, and sensor temperature for at least 30 minutes. A repeated robot trajectory with independent ground truth would then connect sensor-level behavior to odometry and map repeatability.

5. Limitations

The evidence base is small and heterogeneous. Only one peer-reviewed paper reports a full L1-specific odometry error table suitable for secondary analysis. Its Point-LIO reference uses the same LiDAR and IMU, so absolute errors must be interpreted cautiously. Environment-level means are unweighted because sequence lengths are not reported. The field studies evaluate complete robotic systems and cannot isolate sensor error from calibration, motion, software, terrain, or registration choices. The public CERN bag files were not replayed in this report; numerical values were re-analysed from the published tables. The search was targeted rather than systematic, so relevant unpublished or inaccessible tests may exist.

6. Conclusion

The Unitree L1 is best interpreted as a low-cost, wide-field navigation sensor rather than a calibrated survey instrument. Published data show that it can support indoor perception and coarse mapping, but reliable odometry depends strongly on motion, geometric overlap, and processing method. In six CERN environments, registration adapted to non-repetitive scans reduced the unweighted mean trajectory error by more than 90% relative to two generic baselines, while an independent cave study showed that large-scale morphology can be recovered without resolving detailed floor geometry. Functional ROS 2 navigation is demonstrated, but cumulative map drift remains a practical limitation.

The central scientific conclusion is therefore conditional: use the L1 with its IMU and a validated LiDAR-inertial or scan-pattern-aware pipeline; preserve overlap through moderate motion; accumulate scans when detail is needed; and avoid centimetric, real-time, or fine-reconstruction claims until controlled range, coverage, and stream-stability experiments are completed.

Data and reproducibility statement

The Unitree L1 CERN bag files and IMU data are publicly archived under digital object identifier (DOI) 10.5281/zenodo.17091989 [3]. Values used in Figures 1 and 2 and Table 2 were transcribed from Table 3 of Ref. [2]. The calculation rules are stated in Section 2.2, and the six environment-level values are visible in Figure 1. No unpublished Coimbra measurement is included.

References

- [1] Unitree Robotics, Unitree 4D LiDAR-L1 User Manual, version 1.1, June 2024. Available: <https://www.unitree.com/en/download> (accessed 6 August 2026).
- [2] P. Habibiroudkenar, D. Forkel, R. Ojala, S. M. Hari Prasanth, D. H. Davidsson, K. Tammi, E. Matheson, and M. Di Castro, "Non-Repetitive Scan Correction for LiDAR Odometry via Spatial Analysis in CERN Facilities," *IEEE Access*, vol. 14, pp. 52717-52733, 2026, doi: 10.1109/ACCESS.2026.3680632.
- [3] P. Habibiroudkenar, D. Forkel, and D. H. Davidsson, "Unitree Unilidar L1 scans in CERN facilities," *Zenodo*, 2025, doi: 10.5281/zenodo.17091989.
- [4] A. J. Hidding, H. H. Bier, L. Peternel, A. J. Becoy, F. A. P. Romio, and G. Calabrese, "Robotic lava tube mapping and multimodal data collection using quadruped and LiDAR," *Frontiers in Robotics and AI*, vol. 13, art. 1855443, 2026, doi: 10.3389/frobt.2026.1855443.
- [5] A. J. Becoy, K. Khomenko, L. Peternel, and R. T. Rajan, "Autonomous navigation of quadrupeds using coverage path planning with morphological skeleton maps," *Frontiers in Robotics and AI*, vol. 12, art. 1601862, 2025, doi: 10.3389/frobt.2025.1601862.
- [6] B. Muralikrishnan, "Performance Evaluation of Terrestrial Laser Scanners - A Review," *Measurement Science and Technology*, vol. 32, no. 7, art. 072001, 2021, doi: 10.1088/1361-6501/abdae3.
- [7] D. He, W. Xu, N. Chen, F. Kong, C. Yuan, and F. Zhang, "Point-LIO: Robust High-Bandwidth Light Detection and Ranging Inertial Odometry," *Advanced Intelligent Systems*, vol. 5, no. 7, art. 2200459, 2023, doi: 10.1002/aisy.202200459.